GATE-AII BRANCHES
Engineering Mathematics



SINGLE VARIABLE CALCULUS

Lecture No.- 04



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Recap of Previous Lecture



Topic Existence of limits

Topics to be Covered



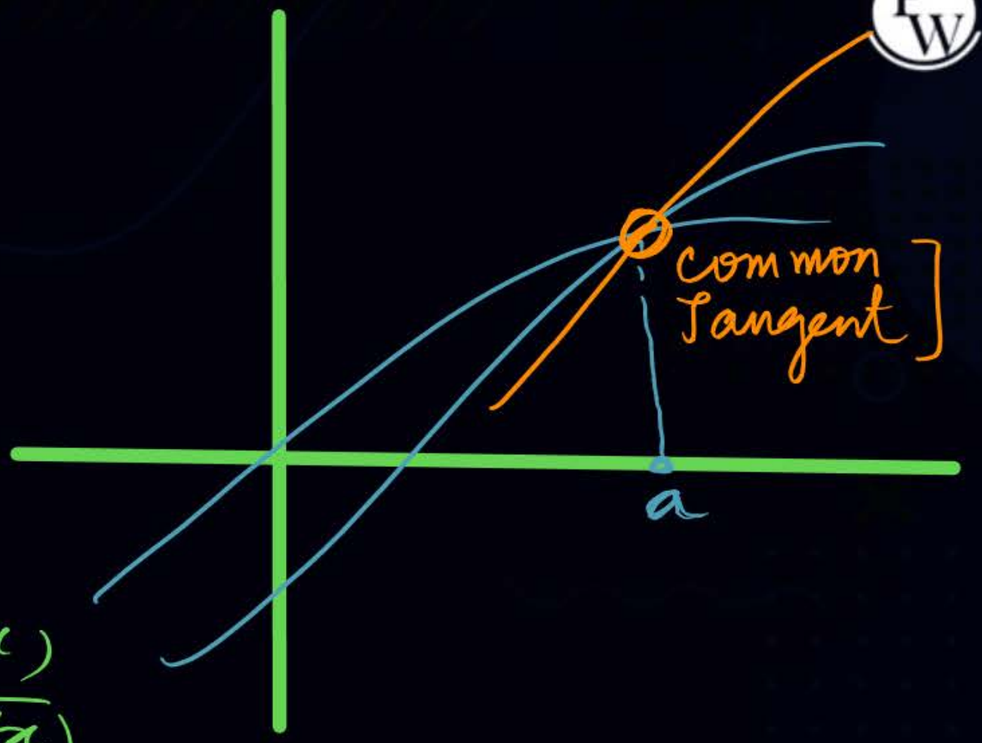
Topic How to evaluate the limits?

L-Hospital Rule:

If $\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}$

applicable for $\frac{\to 0}{\to 0}$ or $\frac{\to \infty}{\to \infty}$

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)} = \lim_{x \to a} \frac{f''(x)}{g''(x)}$$



$\frac{\to 0}{\to 0}$ OR $\frac{\to \infty}{\to \infty}$ → apply L-Hospital → Again $\frac{\to 0}{\to 0}$ OR $\frac{\to \infty}{\to \infty}$

→ remove The $\frac{\to 0}{\to 0}$ OR $\frac{\to \infty}{\to \infty}$

→ Plug Limits Rule

SOME Important Template:

A) $\lim_{x\to 0} \frac{\sin x}{x} = \lim_{x\to 0} \frac{x}{\sin x} = \lim_{x\to 0} \frac{\tan x}{x} = \lim_{x\to 0} \frac{x}{\tan x} = 1$

Using L-Hospital Rule

$$\lim_{x\to 0} \frac{\sin x}{x} = \frac{\to 0}{\to 0} \text{ form} = \lim_{x\to 0} \frac{\cos x}{1} = \cos 0^\circ = 1$$

$$= \lim_{x\to 0} \frac{\frac{d}{dx}(\sin x)}{\frac{d}{dx}(x)}$$

Remove $\frac{\to 0}{\to 0}$ → Plug in Limit

$$\lim_{x\to 0} \frac{x}{\sin x} = \frac{\to 0}{\to 0} = \lim_{x\to 0} \frac{1}{\cos x} = 1$$

$$\lim_{x\to 0} \frac{\tan x}{x} = \frac{\to 0}{\to 0} = \lim_{x\to 0} \frac{\sec^2 x}{1} = 1$$

$$\lim_{x \to 0} \frac{\sin(x)}{x}$$

$$\lim_{f(x) \to 0} \frac{\sin[f(x)]}{f(x)} = 1$$

$$\lim_{f(x) \to 0} \frac{\tan[f(x)]}{f(x)} = 1$$

Another Version

but condition $f(x) \to 0$

condition $f(x) \to 0$

(B) $\lim_{x\to 0}\frac{e^x-1}{x}=1$

$\lim_{x\to 0}\frac{e^x-1}{x}=\frac{\to 0}{\to 0}$ apply L-Hospital Rule

$\Rightarrow \lim_{x\to 0}\frac{\frac{d}{dx}(e^x-1)}{\frac{d}{dx}(x)}=\lim_{x\to 0}\frac{e^x}{1}=e^0=1$

$\frac{d}{dx}(x^n)=nx^{n-1}$

$\frac{d}{dx}(\sin x)=\cos x$

$\frac{d}{dx}(\cos x)=-\sin x$

$\frac{d}{dx}(e^x)=e^x$

C) $\lim_{x\to 0}\frac{a^x-1}{x}=\frac{\to 0}{\to 0}$ form

$=\lim_{x\to 0}\frac{a^x\ln a}{1}=a^0\ln a$

$=\ln a$

remove

$\lim_{x\to 0}\frac{a^x-1}{x}=\ln_e a$

constant No.

$\log \rightarrow$ base 10

$\ln \rightarrow$ base e (Natural Log.)

Topic : Single Variable Calculus

#Q. The value of limit $\lim_{x \to \frac{\pi}{2}}$ (sec x – tan x) is:

$$\lim_{x\to\frac{\pi}{2}}\left[\frac{1}{\cos x}-\frac{\sin x}{\cos x}\right]=\lim_{x\to\frac{\pi}{2}}\left[\frac{1-\sin x}{\cos x}\right]$$

$$\lim_{x\to\frac{\pi}{2}}\left(\frac{1-\sin x}{\cos x}\right)=\frac{\to 0}{\to 0}$$

Using L-Hospital Rule

$$=\lim_{x\to\frac{\pi}{2}}\left[\frac{0-\cos x}{-\sin x}\right]=\lim_{x\to\frac{\pi}{2}}\left[\cot x\right]$$

$$=\cot\frac{\pi}{2}=0$$

A 1

B 0

C -1

D 2

$\sec\frac{\pi}{2}$ Undefined

$\tan\frac{\pi}{2}$ = Undefined

$\lim_{x\to\frac{\pi}{2}}[\sec x-\tan x]$ $\quad \to\infty-\to\infty$ Form

remove – Plug-in

B



Topic : Single Variable Calculus

$\lim_{x\to 0} \frac{\sin(\pi\cos^2 x)}{x^2}$ L-Hospital Rule (A) form $\left(\frac{\to 0}{\to 0}\right)$

Wrong Template

#Q. The value $\lim\limits_{x\to 0} \dfrac{\sin\left(\pi\cos^2 x\right)}{x^2}$ is:

A π

B $-\pi$

C 2π

D -2π

$$\lim_{x\to 0}\frac{\sin(\pi\cos^2 x)}{x^2} = \lim_{x\to 0}\frac{\sin[\pi(1-\sin^2 x)]}{x^2} = \lim_{x\to 0}\frac{\sin(\pi-\pi\sin^2 x)}{x^2}$$

$$= \lim_{x\to 0}\frac{\sin(\pi\sin^2 x)}{x^2}$$

$$= \lim_{x\to 0}\frac{\sin(\pi\sin^2 x)}{\pi\sin^2 x}\cdot\frac{\pi\sin^2 x}{x^2}$$

$$= \lim_{x\to 0}\frac{\sin(\pi\sin^2 x)}{\pi\sin^2 x}\times\pi\times\left(\frac{\sin x}{x}\right)^2$$

$$= 1\times\pi\times 1$$

$$= \pi$$

$\sin(\pi-\theta)$

$(\pi-\theta)$

$(+,+)$

$$\lim_{x\to 0}\frac{\sin[f(x)]}{f(x)} = 1$$

L-Hospital Rule:

$$\lim_{x\to 0}\frac{\sin(\pi\cos^2 x)}{x^2}$$

Do yourself



Topic : Single Variable Calculus

#Q. The limiting value of $x \sin \frac{1}{x}$ as $x \to 0$ is:

$= \lim_{x \to 0} x \sin\left(\frac{1}{x}\right)$

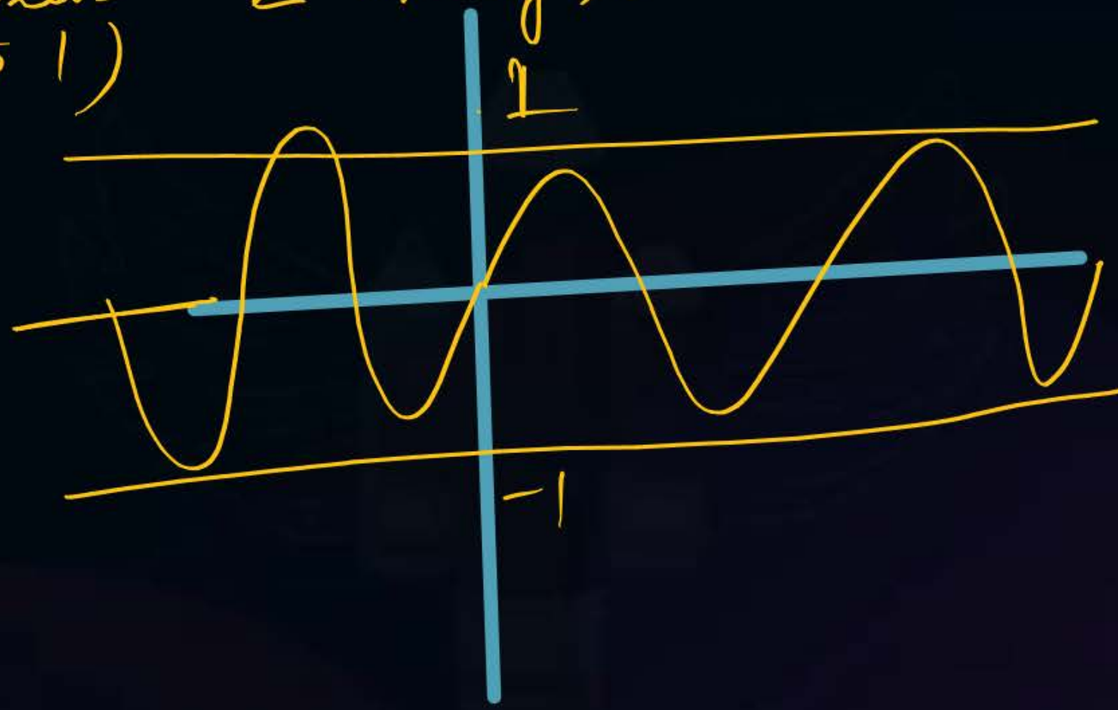
$= 0 \times$ (oscillation between -1 to 1)

$= 0$

$\lim_{x \to 0} x \sin\left(\frac{1}{x}\right)$

$= \lim_{x \to 0} \frac{\sin\left(\frac{1}{x}\right)}{\left(\frac{1}{x}\right)}$ $= \sin\infty = -1$ to 1 Oscillate

Wrong ✗



A. 1

B. 0 ✓

C. -1

D. Limit does not exist



Topic : Single Variable Calculus

#Q. Evaluate : $\lim_{x\to 1} \frac{2^x - 2}{x-1}$ $\Rightarrow \frac{\to 0}{\to 0}$

L-Hospital Rule Template

$x - 1 = t$

$\Rightarrow \lim_{x\to 1} 2\left(\frac{2^{x-1}-1}{(x-1)}\right)$

$\lim_{t\to 0} 2\left(\frac{2^t - 1}{t}\right)$

$\Rightarrow 2\ln 2 = \ln 2^2 = \ln 4$

$\lim_{x\to 0} \frac{a^x - 1}{x} = \ln a$

$= \lim_{t\to 0} \frac{(2^t - 1)}{t} 2$

$= 2\ln 2$

IInd method $= \lim_{x\to 1} \frac{2^x - 2}{(x-1)} = \lim_{x\to 1} \frac{2^x \ln 2}{1}$

$= 2^1 \ln 2 = 2\ln 2$



Topic : Single Variable Calculus

#Q. Evaluate : $\lim_{x \to a} \dfrac{e^{\sqrt{x}} - e^{\sqrt{a}}}{x - a}$

$$\Rightarrow \lim_{x \to a} \frac{e^{\sqrt{a}} \left[e^{\sqrt{x} - \sqrt{a}} - 1 \right)}{(\sqrt{x} - \sqrt{a})(\sqrt{x} + \sqrt{a})}$$

$$= \lim_{x \to a} e^{\sqrt{a}} \left[\frac{\dfrac{e^{\sqrt{x} - \sqrt{a}} - 1}{\sqrt{x} - \sqrt{a}}}{\sqrt{x} + \sqrt{a}} \right]$$

$$= \frac{e^{\sqrt{a}}}{\sqrt{a} + \sqrt{a}} = \frac{e^{\sqrt{a}}}{2\sqrt{a}}$$

$$\lim_{x \to a} \frac{e^{\sqrt{x}} \cdot \frac{1}{2\sqrt{x}} - 0}{1}$$

$$= \frac{e^{\sqrt{a}}}{2\sqrt{a}}$$

ANSWER



Topic : Single Variable Calculus

#Q. The value of $\lim\limits_{x \to 0} \dfrac{e^{\tan x} - e^{x}}{\tan x - x}$ is:

$$\lim_{x \to 0} e^{x} \left[\frac{e^{\tan x - x} - 1}{\tan x - x} \right]$$

$= e^{0}$

$= 1$

A 1

B 2

C 1/2

D 0



Topic : Single Variable Calculus

#Q. Evaluate: $\lim_{x\to a}\frac{x^a-a^x}{x^x-a^a}$ $\left(\frac{0}{0}\text{ type of indeterminate form}\right)$

Using L-Hospital Rule

$$\lim_{x\to a}\frac{x^a-a^x}{x^x-a^a}=\frac{\to 0}{\to 0}$$

$$\lim_{x\to a}\frac{\frac{d}{dx}(x^a-a^x)}{\frac{d}{dx}x^x-\frac{d}{dx}a^a}$$

$$=\lim_{x\to a}\frac{ax^{a-1}-a^x\ln a}{x^x(1+\ln x)}=\frac{aa^{a-1}-a^a\ln a}{a^a(1+\ln a)}=\left(\frac{1-\ln a}{1+\ln a}\right)$$ Ans

✓ $\frac{d}{dx}(x^n)=nx^{n-1}$

$\frac{d}{dx}(x^a)=ax^{a-1}$

✓ $\frac{d}{dx}(a^x)=a^x\ln a$

✓ $\frac{d}{dx}(a^a)=0$

$y = x^x$ w.r.t to x

$\log y = x \times \log x$

I II

w.r.t to x

$$\frac{1}{y}\frac{dy}{dx} = x \cdot \frac{d}{dx}(\log x) + \log x \cdot \frac{d}{dx}(x)$$

$$\frac{dy}{dx} = y\left[x \cdot \frac{1}{x} + \log x \cdot 1\right]$$

$$\frac{dy}{dx} = x^x[1 + \log x]$$

$$= x x^{x-1} + x^x \ln x$$

$y = [\text{function}]^{\text{function}}$

both sides Log.

$$\frac{d}{dx}(I \times II) = I \times \frac{d}{dx}(II) + II \frac{d}{dx}(I)$$

$f(x) = x^x = x$ — HEART

$= x x^{x-1}$

mann

II $\longrightarrow$ $x^x \longrightarrow a^x$

a

$= x^x \ln x$

$$f(x) = (\sin x)^{\frac{\cos x}{n}}$$

$$= \cos x (\sin x)^{\cos x - 1} + (\sin x)^{\cos x} \log \sin x$$



Topic : Single Variable Calculus

#Q. **Evaluate :** $\lim_{x\to 0} \dfrac{x(e^x-1)+2(\cos x-1)}{x(1-\cos x)}$

$= \dfrac{\to 0}{\to 0}$

$= \dfrac{\to 0}{\to 0}$

$= \dfrac{\to 0}{\to 0}$

3 times Using L-Hospital Rule

DO yourself in your NEST

Ans = 3



Topic : Single Variable Calculus

GATE

#Q. Evaluate :

$\lim_{\theta \to 0} \frac{\sin m\theta}{\theta}$, where m is an integer, is one of the following:

$\lim_{\theta \to 0} \frac{\sin m\theta}{\theta} \times \frac{m}{m}$

$= m \lim_{\theta \to 0} \frac{\sin m\theta}{m\theta}$

$= 1$

A m

B $m\pi$

C $m\theta$

D 1



Topic : Single Variable Calculus

#Q. Evaluate : $\lim_{x \to 0} \left(\frac{1-\cos x}{x^2} \right)$ is

$= \frac{1}{2}$

Using L-Hospital Rule

$= \lim_{x \to 0} \frac{0 + \sin x}{2x}$

$= \frac{1}{2} \lim_{x \to 0} \frac{\sin x}{x}$

$= \frac{1}{2}$



Topic : Single Variable Calculus

#Q. Evaluate : $\lim_{x\to 0} \frac{x-\sin x}{1-\cos x}$ is

Using L-Hospital Rule :

$$= \lim_{x\to 0} \frac{1-\cos x}{0+\sin x}$$

$$= \lim_{x\to 0} \frac{1-\cos x}{\sin x} = \frac{\to 0}{\to 0}$$

$$= \text{Again}$$

$$= \lim_{x\to 0} \frac{0+\sin x}{\cos x}$$

$$= \lim_{x\to 0} \tan x = 0$$



Topic : Single Variable Calculus

#Q. Evaluate : $\lim_{x\to 0}\left(\frac{e^{2x}-1}{\sin(4x)}\right)$ Using L-Hospital Rule

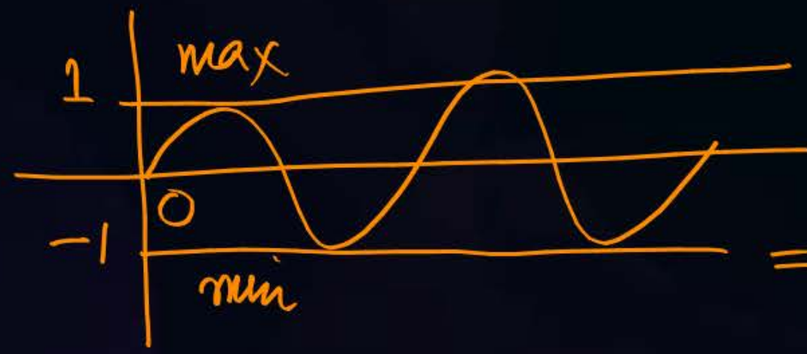
$$\Rightarrow \lim_{x\to 0}\frac{e^{2x}-1}{\sin 4x} = \frac{2}{4} = \frac{1}{2}$$

Topic : Single Variable Calculus

#Q. Evaluate : $\lim_{x\to\infty}\left(\frac{x+\sin x}{x}\right)$ is equal to

$$\lim_{x\to\infty}\left[\frac{1+\frac{\sin x}{x}}{1}\right] = 1$$

$$\lim_{x\to\infty}\frac{\sin x}{x} = 0$$



$$\lim_{x\to\infty}\frac{[-1,1]}{x} = 0$$

A $-\infty$

B 0

C 1

D ∞

Topic : Single Variable Calculus

#Q. The expression $\lim\limits_{a\to 0}\dfrac{x^a-1}{a}$ is equal to

(A) $\log x$

(B) 0

(C) $x\log x$

(D) ∞

$a \to 0$ variable

variable $= a$

constant $= x$

$= \lim\limits_{a\to 0}\dfrac{x^a \log x}{1}$

$= x^0 \log x$

$= \log x$

$= \lim\limits_{a\to 0}\dfrac{\frac{d}{da}(x^a-1)}{\frac{d}{da}(a)}$

$x^a \log x$

$x \to 0$ variable $\to$ constant

a variable

(constant)^{variable}

x



Topic : Single Variable Calculus

#Q. Evaluate :

The value of $\lim_{x\to 0} \frac{1-\cos\left(x^2\right)}{2x^4}$ is

H.W

Do yomself

A 0

B 1/2

C 1/4

D undefined



Topic : Single Variable Calculus

#Q. Evaluate :

$$\lim_{x\to 0}\left(\frac{-\sin x}{2\sin x + x\cos x}\right) \text{ is } _____.$$

H.W

Do yourself



Topic : Single Variable Calculus

#Q. Evaluate :

$$\lim_{x\to 0}\left(\frac{\tan x}{x^2 - x}\right) \text{ is } _____.$$

H.W

Do yourself



Topic : Single Variable Calculus

H.W

Do yourself

#Q. The value of $\lim_{x\to 1} \frac{x^7 - 2x^5 + 1}{x^3 - 3x^2 + 2}$

A is 0

B is – 1

C is 1

D Does not exit



Topic : Single Variable Calculus

H.W

Do yourself

#Q. Compute $\lim_{x\to 3} \frac{x^4-81}{2x^2-5x-3}$

A 1

B limit does not exit

C 53/12

D 108/7


THANK - YOU